

FORENSIC ANALYSIS REPORT

Image: jo-favorites-usb-2009-12-11.E01 • 2026-02-22 20:18:12 • Model: meta-llama/llama-3.1-8b-instruct

Risk Level: MEDIUM

EXECUTIVE SUMMARY

The system's timestamp has been manipulated, and the Volume Shadow Copy Service has been deleted, indicating potential anti-forensic activity.

Analysis completed in 8.8 seconds. AI analysis identified 3 findings. Anti-forensic analysis detected 260 potential indicators.

AI ANALYSIS FINDINGS

Finding #1: timestomping	MEDIUM	80% conf
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Evidence: Future timestamps found: 1, USN journal missing

Explanation: The presence of future timestamps and a missing USN journal suggests that the system's timestamp has been manipulated.

Recommendation: Verify the system's timestamp and investigate the files with future timestamps for potential evidence of tampering.

Finding #2: shadow_deletion	HIGH	90% conf
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Evidence: \$Extend metadata does not exist, System Volume Information folder is missing

Explanation: The absence of the \$Extend metadata and the System Volume Information folder indicates that the Volume Shadow Copy Service has been deleted.

Recommendation: Investigate the system's event logs for any signs of VSS deletion and examine the system's registry for any suspicious changes.

Finding #3: deletion	LOW	40% conf
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Evidence: Orphaned files: 1

Explanation: The presence of an orphaned file suggests that a file has been deleted, but the system's MFT entry has not been properly updated.

Recommendation: Verify the file's deletion and investigate the system's event logs for any signs of file deletion.

ANTI-FORENSIC ANALYSIS

TIMESTOMPING DETECTION [256 indicators]

Detects manipulation of file timestamps

- Partition 0: Zero timestamp detected: r/r 3: JO FAVORITE (Volume Label Entry) 2009-11-20 09:30:34 (IST) 0000-00-00 00:
 - Partition 0: Zero timestamp detected: r/r 5: ._Trashes 2009-11-17 16:45:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: r/r 6: DSC00009.JPG 2009-11-11 09:16:56 (IST) 2009-11-24 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: d/d 8: .Trashes 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-00 0
 - Partition 0: Zero timestamp detected: + d/d * 133: ._02 2009-11-20 09:25:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: + r/r * 135: ._502 2009-11-20 09:25:28 (IST) 2009-11-20 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: + r/r * 136: ._502~1 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: + r/r * 138: ._501 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: d/d * 10: .fsevents 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00
 - Partition 0: Zero timestamp detected: d/d 13: .Spotlight-V100 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000
 - Partition 0: Zero timestamp detected: + d/d 646: Store-V1 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: ++ r/r 775: VolumeConfig.plist 2009-11-17 16:45:28 (IST) 2009-11-20 00:00:00 (IS
 - Partition 0: Zero timestamp detected: ++ d/d 777: Stores 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: +++ d/d 1544: BC650BA8-C5B5-4A64-8BDB-18A9533BF1D7 2009-11-17 16:45:28 (IST) 200
 - Partition 0: Zero timestamp detected: ++++ r/r 1669: psid.db 2009-11-20 09:30:34 (IST) 2009-11-20 00:00:00 (IST) 0000-
- ... and 241 more

SHADOW COPY DELETION [Clear]

Evidence of VSS deletion attempts

No indicators found in this category.

HIDDEN DATA STREAMS (ADS) [Clear]

Alternate Data Streams detection

No indicators found in this category.

LOG CLEARING EVIDENCE [2 indicators]

Evidence of log file clearing

- Partition 0: Few or no log files found - possible clearing
- Partition 0: No Windows event log files (.evtx) found

REGISTRY TAMPERING [Clear]

Registry modification indicators

No indicators found in this category.

FILE DELETION TRACKING [1 indicator]

Deleted file recovery opportunities

- Partition 0: Orphaned files found (deleted but recoverable): 1 entries

MFT ANOMALIES [1 indicator]

Master File Table irregularities

- Partition 0: MFT contains 4 system file entries

SUMMARY STATISTICS

Category	Indicators
Timestamping	256
Shadow Copy Deletion	0
Hidden Streams (ADS)	0
Log Clearing	2
Registry Tampering	0
File Deletion	1
MFT Anomalies	1
TOTAL	260

Overall Anti-Forensic Risk Level: HIGH

RECOMMENDATIONS

1. Verify the system's timestamp and investigate the files with future timestamps for potential evidence of tampering.
2. Investigate the system's event logs for any signs of VSS deletion and examine the system's registry for any potential evidence of tampering.

Additional Anti-Forensic Investigation Steps:

- Conduct detailed timeline analysis around detected anomalies.

- Examine Windows Event Logs for evidence of vssadmin or wevtutil commands.
- Analyze \$MFT and \$UsnJrnl for file system manipulation evidence.
- Review Registry hives for persistence mechanisms and tool artifacts.
- Search for known anti-forensic tool signatures and artifacts.